In India, startups are increasingly using open source IoT solutions to make advanced technologies more affordable and accessible. A prominent example is IoT OpenLab, an initiative by the Software Technology Parks of India (STPI) under the Ministry of Electronics and Information Technology (MeitY), Government of India. This lab supports startups by providing infrastructure, tools, and mentorship, focusing on sectors like agriculture and healthcare. It also aligns with national initiatives like Digital India and Make-in-India, helping reduce electronics imports and promote local innovation. Another example is iSPIRT's Open Source Healthcare IoT, which utilises open frameworks like the Bharat Health Stack to develop connected healthcare solutions. These solutions improve communication and data sharing within healthcare systems, leading to better diagnostics and overall efficiency. By using open standards, the Bharat Health Stack ensures health data can be securely shared across various platforms. This makes it easier for doctors, clinics, and hospitals to access and update patient records in real time. For instance, a patient's medical history can be made available to all their healthcare providers to ensure timely and accurate care. This open approach also reduces costs, making healthcare more affordable and accessible.

Globally, open source IoT is making waves as well. Arduino, an open source electronics platform, is widely used for building custom IoT projects—from smart weather stations to home security systems—empowering hobbyists and professionals alike.

In India and around the world, open source IoT is helping individuals and communities create their own solutions to solve local problems. This is leading to new ideas, saving money, and making technology available to everyone.

A brief history of open source and IoT

Open source technology became popular in the 1990s when developers started sharing their code to accelerate innovation. A prominent example is Linux, which showed how developers across the world could work together to build something powerful and free to use.

The idea of the Internet of Things (IoT) is relatively recent. The term IoT was coined in 1999 by Kevin Ashton, a British technology pioneer, but the concept of connected devices dates back even earlier. In the 1980s, engineers at Carnegie Mellon University connected a Coca-Cola vending machine to a local network, allowing them to remotely monitor the machine's status, including the temperature of the drinks. This early experiment demonstrated the potential of remotely controlling and monitoring devices.

However, long before the term IoT came into use, the concept of embedded systems was already gaining ground. Embedded systems were the technologies used in everyday devices, such as washing machines, microwaves, and vending machines to make them work in an automated way. These systems were 'embedded' within products to make them smart and efficient. As the need for connecting these devices to the internet grew, what was once simply called embedded systems evolved into what we now know as IoT. In simpler terms, IoT took the old idea of embedded systems and added the power of connectivity, enabling devices to communicate with one another over the internet.

The real growth of IoT started in the 2010s when things like faster internet, affordable sensors, and cloud computing made it much easier to connect devices. As IoT grew, the demand for tools to manage and develop these systems also increased, and this is where open source technology stepped in to provide affordable, flexible solutions.

How open source and IoT work together

IoT systems can be quite complex, as they involve hardware like sensors, software, and data. Building these systems from scratch can be expensive and time-consuming. However, open source technologies have made it easier and more affordable to create and deploy IoT systems without requiring advanced technical knowledge or large investments.

For example, consider a farmer in India who wants to monitor soil moisture levels and automate the watering of crops when the soil dries out. Instead of building an expensive system from scratch, the farmer can use ThingSpeak, an open source platform that makes it easy to collect and analyse data from sensors. ThingSpeak allows the farmer to track soil moisture in real-time and determine when watering is needed.

Next, the farmer can use Node-RED, another open source tool, to automate the watering process. Node-RED is a no-code platform, meaning the farmer doesn't need to know how to program. With Node-RED, the farmer can simply drag and drop blocks to create a workflow that triggers the irrigation system. For instance, when the soil moisture falls below a certain level, Node-RED can automatically activate the irrigation systems, ensuring the crops get watered without any manual effort. By using ThingSpeak and Node-RED together, even small-scale farmers can benefit from smart technology. This solution improves farming efficiency, saves water, and boosts crop yields—all at a much lower cost than traditional systems.

Let's look at another example. Suppose you run a bakery and want to cut electricity costs. You decide to automate the cooling system in your storage room, where you keep perishable items like butter and cream. Using a temperature sensor and a Raspberry Pi (a low-cost, open source